

Capstone Project Proposal

Analyzing Adverse Drug Event Reports from OpenFDA

1. Introduction and Problem Statement

Adverse Drug Events (ADEs) are unwanted symptoms or problems that happen while a patient is taking medicine. These can be mild side effects or serious issues like needing to go to the hospital or even death.

To keep track of drug safety, the U.S. Food and Drug Administration (FDA) runs the Adverse Event Reporting System (FAERS). This is a central database that gathers reports from doctors, patients, and drug companies.

Even though this database has a lot of valuable information, it is not used enough because the data is complex and not always easy to work with. This project will collect and study these reports to answer important questions such as:

- Which drugs cause the most serious problems like hospital visits or death?
- What kinds of side effects happen most often?
- Are certain groups of people, like different ages or genders, more likely to experience specific problems?
- Can we find signs of drug safety issues that have not been noticed before?

2. Context and Use Case

[openFDA](#) is making it easier for researchers, scientists, web developers, and other FDA regulatory stakeholders to access and use those datasets in an open standard format.

The project aims to create easy access to public data and a new level of openness and accountability, ensure the privacy and security of public FDA data, educate the public, and save lives. Members of the scientific community can use openFDA to have their applications automatically query the data through application program interfaces (APIs). OpenFDA increases the efficiency and speed of accessing datasets by using cutting-edge, open-source code modules in a cloud-based environment.

3. Data Source

The dataset comes from the **OpenFDA API**, which provides public access to the **FAERS** data.

Dataset Features:

- Drug names (brand & generic)

- Patient demographics (age, gender, weight)
- Reported outcomes (hospitalization, death, life-threatening events)
- Dates of adverse event reports
- Seriousness categorization
- Source of report (healthcare provider, consumer, manufacturer)

4. Data Acquisition

- Via OpenFDA REST API
- Option to download bulk data in JSON format

5. Data Preparation Plan

Expected Tasks:

- **Extraction:**
 - Use API or bulk file downloads to collect event data
- **Flattening:**
 - Convert nested JSON to tabular format
- **Cleaning:**
 - Handle missing fields
 - Standardize drug names (normalize brand vs. generic)
 - Remove duplicates or duplicate reports
- **Joining:**
 - Link patient data with drug and outcome fields
- **Transformation:**
 - Convert date strings for time-series analysis
 - Categorize age into bins

6. Tools and Technologies

Component	Tools / Libraries
Data Extraction	Python (requests, json)
Data Transformation	Python (pandas, numpy)
Storage	PostgreSQL
Data Analysis	SQL, Python
Visualization	Tableau / matplotlib, plotly

Optional Orchestration

Apache Airflow (for automated ETL)

Optional Cloud Deployment

AWS EC2 + S3 or GCP Cloud Storage
or Azure blob storage